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A Cubic Separation Law for Wide Binaries: an Exponent With No Free Parameters

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*Version 6 · 2026-08-01. (Zenodo v4 and v5 were republications — a reproducibility-file restoration and the \$ 2kernelnotice—thatleftthepapertextatitsVersion3state, sothisisthefirsttextrevisionsince2026–07–30.NumberingfollowstheZenodorecord.)**ThisversionDEMOTESThepaper'sresultfromapredictiontoaCC*
(i)Theeigenvectoridentity" $\square_u u_\mu = -\Omega^2 u_\mu$ " asserted by v2 — v3 is **false as stated — — it was verified on a two — component spatial * vector; $\square_u$ acts block-diagonally, annihilating the time leg and returning $-(\gamma\Omega)^2$ on the spatial legs, so u_μ is not an eigenvector. (ii) The argument for why a gate must exist relied on " $|K| = 1$ exactly", which is a property of the **retired** $\alpha = 1$ kernel; the conclusion survives on $\alpha = 2$ but the exactness does not. (iii) **The branch selection §2 made is WITHDRAWN:** the frequency ("AC") reading is the one under which the action's modification is amplitude-free and the framework has no rotation curves, so the phenomenology runs on the first-moment closure, whose argument is $+|a|^2$ — and a low-pass passes DC at unit gain, so γ_v reverts to the ungated ≈ 1.02 . That is **this paper's own falsifier #5, reached from the framework's own side.** The s^3 law is therefore retained as a **conditional** result — valid on the AC branch, which this author's own analysis no longer supports — not as the framework's prediction. Nothing measured changes. Scripts: `mi_dcac_branch_verdict_withdrawn_2026.py`, `mi_alpha1_exclusive_class_audit_2026.py`.**

Version 3 · 2026-07-30. The s^3 prediction, its five falsifiers and the frozen pre-registration are UNCHANGED. v3 (i) reduces the coefficient to a single rational, $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ with $\kappa = 1/2$ (§1); (ii) tightens the ECCENTRIC-ORBIT error budget from a factor of several to $\lesssim 8\%$ using the action's own $\square_u$, and records two exact identities (§2.1); (iii) adds an independent CMB check of the a_0 footing, crediting Gnedin (2008) for posing and first deciding that fork (§8); and (iv) reports six closed classes in the search for what fixes κ — a negative result. Version 2 (2026-07-28) settled the branch ambiguity, generalised the exponent, and resolved feasibility negatively; Version 1 (2026-07-25) is superseded. Scripts listed in §8.

The claim, first — and what it is now conditional on

Read this before the equation. What follows is a **conditional** result, not this framework’s prediction. It holds if the kernel responds to the orbital **frequency** (the “AC” branch). Versions 2–3 claimed the action settles that question in favour of AC; **v6 withdraws that claim** (§2), because the AC reading is the one in which the action’s modification is amplitude-free and the framework has no rotation curves. On the branch the framework’s phenomenology actually uses, the gate never suppresses anything and $\gamma_v \approx 1.02$ inside 30 kAU — **which is falsifier #5 below, reached from the framework’s own side rather than from data.** The derivation is kept on the record because it is correct given its branch, and because the exponent-3 signature remains the sharp test if that branch is ever revived. It should not be cited as “the framework predicts s^3 ”.

On the AC branch, in wide binaries wider than about 50 kAU, the relative velocity should exceed the Newtonian value by an amount growing as the **cube of the separation**:

$$\gamma_v(s) - 1 \simeq \frac{1}{2}[\nu(y_{\text{ext}}) - 1] \frac{\omega_c^2}{GM} s^3.$$

The **exponent is 3**. Given the branch it is derived, not fitted, and it cannot be tuned. Every known contamination channel in wide-binary samples rises as $s^{1/2}$ instead. Exponent 3 against exponent 0.5 — that contrast is what the measurement would be.

What would kill it. Any one of: no excess beyond ~50 kAU in a contamination-controlled sample; an excess whose log-slope is inconsistent with 3; a knee that does not move as $M^{1/3}$; an amplitude inconsistent with ω_c^2/GM ; or a boost $\gamma_v \approx 1.02$ found inside 30 kAU (**renumbered 2026-07-30 from 1.09 under the retired $\alpha = 1$ kernel; see the notice in §2**). As of v6 the last of these is no longer only a data risk — the framework’s own DC branch predicts it, so on that branch this law is already falsified without any new observation.

What it cannot do. It cannot be measured with Gaia DR3. We checked the catalogue ourselves; see §6.

1. Two scales, one of them free

The framework is modified inertia — the modification is in the matter sector, not in gravity. Its acceleration scale is tied to the cosmological constant:

$$a_0 = \frac{cH_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = 9.36 \times 10^{-11} \text{ m s}^{-2}, \quad Z = \sqrt{32\pi/3} = 5.78881.$$

New in v3 — the coefficient is a single rational, and “ $32\pi/3$ ” is an artifact. Writing 32π as $8\pi/\kappa^2$ and substituting $\Lambda = 8\pi G\rho_\Lambda/c^2$, every π , the 32 and the 3 all cancel:

$$\boxed{a_0 = \kappa c \sqrt{G\rho_\Lambda}}, \quad \kappa = \frac{1}{2}.$$

So $32\pi/3$ is not a compound of three geometric factors — it is what one statement looks like after being routed through Λ and Einstein’s 8π . The framework’s entire dimensionless content is the single **rational** $\kappa = 1/2$, and the honest form of the open problem is why $\kappa = \frac{1}{2}$, not why $32\pi/3$.

Two consequences worth recording. (i) The standing transcendence obstruction — “ Z carries a transcendental $\sqrt{\pi}$ while flavour data are algebraic” — is correct about Z , but that $\sqrt{\pi}$ belongs to **general relativity’s** 8π , not to this framework, whose own number is rational. (ii) κ remains **postulated**: an exhaustive sweep of the kernel’s own conditions (Herglotz sum rule, $K(0) = 0$, $\|K\| \leq 1$, retarded analyticity, branch point, thermal saturation against every forced kernel constant, and the location statistics of the spectral measure) found **none** that fixes it, because each is imposed at a scale-free point of the dimensionless axis $z = \square_u/a_0^2$ and is therefore invariant under $a_0 \rightarrow \lambda a_0$. Six classes closed; the value is not derived.

KERNEL CHANGED 2026-07-30. This paper was written with $\nu(y) = \sqrt{1+1/y}$ ($\alpha = 1$). The framework has since adopted $\mu(x) = x/\sqrt{1+x^2}$ ($\alpha = 2$, Milgrom 1983), because the $\alpha = 1$ tail implies a constant sunward anomaly $a_0/2$ that is **1279× the Earth 2σ ephemeris bound**. Two consequences for this paper, both stated rather than absorbed: (i) the identity $g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}}$ below is an $\alpha = 1$ identity and no longer holds — it is retained as attribution, not as a live relation; (ii) the ungated boost quoted throughout as $\gamma_v \approx 1.09$ becomes ≈ 1.02 (orientation-averaged 1.0246, range 1.0182–1.0350 over both a_0 footings and both frozen g_{ext} values), per Amendment 3 to PREREGISTRATION_DR4.md, filed in the open before DR4. The cubic gate law, its $s^{1/2}$ -versus- s^3 contrast and the $M^{1/3}$ knee are properties of the **gate**, not of the tail exponent, so they are untouched by the kernel change specifically. **v6: this paragraph’s original closing claim — “the paper’s central result is unaffected ... the gated prediction remains $\gamma_v \rightarrow 1.000$ ” — is RETRACTED.** It is unaffected by the $\alpha = 1 \rightarrow \alpha = 2$ switch, but not by the separate withdrawal of the AC branch in §2, on which the gated value $\gamma_v \rightarrow 1.000$ entirely depends. On the framework’s own DC branch there is no gate suppression and $\gamma_v \approx 1.02$.

Credit, plainly stated. The interpolation $\nu(y) = \sqrt{1+1/y}$ and the identity $g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}}$ are Milgrom (1999, Phys. Lett. A **253**, 273), Eqs. (8)–(9), with coefficient $2cH_\Lambda$. Milgrom (1999, 2015) and Smolin (2017, Phys. Rev. D **96**, 083523) had already tied a_0 to Λ with a 2π coefficient. **No priority is claimed for the Λ tie.** What is ours: the coefficient cH_Λ/Z , the modified-inertia completion, and the prediction below.

The second scale is a response gate. The matter sector’s effective response is

$$K_{\text{eff}} = 1 - S(|a|/a_0) G(\omega), \quad G(\omega) = \frac{1}{1 + i\omega/\omega_c}, \quad S \rightarrow \frac{a_0}{2g_N} \quad (\text{deep-Newtonian}),$$

with $\omega_c \in [1.782, 2.211] \times 10^{-14} \text{ s}^{-1}$. Every number below is carried on both cosmological footings ($a_0 = cH_\Lambda/Z$ and $a_0 = cH_0/Z$, differing by $\sim 21\%$).

Why a gate must exist — on the AC branch. On the frequency axis the kernel saturates and has no high-frequency roll-off, so K gives an inner-disk star and the Earth the **same** response. Since the solar system must be Newtonian and galaxies must not be, something frequency-dependent with a roll-off is required. On that branch the gate’s existence is forced, not assumed.

v6 — and this is why the gate is an AC-branch object specifically. The argument above needs a reading in which K cannot already tell the Earth from an inner-disk star. That is true on the frequency axis, where both saturate. It is **false** on the branch the framework’s phenomenology actually uses: there K ’s argument is the acceleration magnitude $|a|^2/a_0^2$, and the Earth ($|a|/a_0 \sim 10^7$, so $\mu \rightarrow 1$, Newtonian) is separated from a galactic orbit ($|a|/a_0 \sim 1$) **by the argument itself** — which is ordinary MOND logic and needs no gate. So ω_c is not a forced ingredient of the

framework; it is a repair required only by the AC reading, and the AC reading is the one withdrawn in §2. Read together with §1's " ω_c is a free parameter", the honest position is that the gate is **neither forced nor anchored**.

Corrected in v6 — the word exactly was $\alpha = 1$ -specific. Versions 1–3 argued this from " $|K| = 1$ exactly for every $\omega > a_0/2c$ ". That is a property of the **retired** $\alpha = 1$ kernel, which is unimodular on the whole of its cut $z \leq -\frac{1}{4}$. The $\alpha = 2$ kernel now in force is not: with $w = \omega c/a_0$ it gives $|K_2| = (1 - w^{-2})^{-1/2}$, which equals $\sqrt{2}$ at $w = \sqrt{2}$ — well inside "every $\omega > a_0/2c$ " — and is within 1% of unity only for $w > 7.1$. **The conclusion is unaffected.** The two systems the argument compares sit at $w = 1.7 \times 10^4$ (inner-disk star) and 6.3×10^{11} (Earth) on both footings, where $|K_2|$ departs from 1 by at most 1.7×10^{-9} : the responses really are the same, and a roll-off really is required. Only the claim of exactness changes. `mi_alpha1_exclusive_class_audit_2026.py` (25/25).

Why its scale is not. Stated against interest: ω_c is a **free parameter**. A dimensional census puts every intrinsic scale of the theory 3.2–5.6 decades away from it, and consistency alone (galaxies must survive, the solar-system monopole must not) brackets it only to ~ 3 orders of magnitude. So a_0 's value, Z , the sign $s = -1$, **and** ω_c are postulated, not derived. Version 1 did not say this clearly enough; it does now.

2. What frequency a binary presents to the gate

The kernel's argument is fixed by the action, not chosen. The committed matter term is

$$S_{\text{matter}} = -\frac{1}{2} \int d^4x \sqrt{-g} \rho_m [s u^\mu K(\square_u/a_0^2) u_\mu], \quad \square_u f = u^a \nabla_a (u^b \nabla_b f),$$

so $\square_u$ is the second derivative along the worldline. On a circular orbit it acts **block-diagonally** on u_μ : it annihilates the time leg and returns $-(\gamma\Omega)^2$ on the spatial legs,

$$\square_u u^0 = 0, \quad \square_u u^i = -(\gamma\Omega)^2 u^i \quad (\text{exactly}).$$

Because $K(0) = 0$ the time eigenspace drops out, and the contraction reduces to $u^\mu K(\square_u/a_0^2) u_\mu = \gamma^2 v^2 K(-(\gamma\Omega c/a_0)^2)$. The argument is therefore $z = -(\gamma\Omega c/a_0)^2$ — the **orbital frequency**. The acceleration magnitude $|a| = \gamma^2 \Omega^2 R$ genuinely is constant on a circular orbit, but this contraction does not feed $|a|$ to K .

CORRECTED AND PARTLY WITHDRAWN IN v6 — READ BEFORE USING §3. Versions 2 and 3 asserted that " u_μ is an eigenvector" with " $\square_u u_\mu = -\Omega^2 u_\mu$ identically". **That is false as stated:** it was verified on a two-component spatial vector, with the time leg never formed. On the actual 4-velocity the two eigenvalues differ (0 and $-(\gamma\Omega)^2$), so u_μ is not an eigenvector, and pulling K out of the contraction as $K(z)(u \cdot u) = -K(z)$ is invalid — the correct block value $+\gamma^2 v^2 K(z)$ differs in sign and by $(v/c)^2$.

More seriously, the branch selection this section made is WITHDRAWN. Choosing the frequency ("AC") reading by reading $\square_u$ off the action selects the one reading under which the action's modification is amplitude-free — $|K| \rightarrow 1$ at every

real system, on both kernels and both footings, against a law requiring $\mu_{\text{fw}}(1) = 0.618$ — i.e. the reading in which the framework has **no rotation curves at all**. The framework’s phenomenology runs instead on the first-moment closure, whose argument is $+|a|^2 \geq 0$ (Theorem B). A one-pole low-pass passes DC at **unit gain whatever** ω_c **is**, and $|a|$ is exactly constant on a circle, so the $\text{Re } G = 0.005\text{--}0.008$ suppression used in §3 **does not apply** on that branch and γ_v reverts to the ungated ≈ 1.02 ($\alpha = 2$).

Consequence, stated against interest: that is this paper’s own falsifier #5 (“a boost $\gamma_v \approx 1.02$ found inside 30 kAU”), reached from the framework’s own side rather than from data. The s^3 law of §4 is therefore **conditional on the AC branch**, and the AC branch is no longer the one this author’s own analysis supports. §§3–5 must be read as conditional, not as the framework’s prediction. The sentence “The result below no longer rests on a choice” is retracted: it does rest on a choice, and the choice now appears to go the other way.

Sources: `mi_dcac_branch_verdict_withdrawn_2026.py` (29/29), `mi_action_eom_vs_rar_2026.py`, `mi_dcac_split_settled_2026.py`. What is not affected: Theorem B itself, the exact block form above, and every measurement — the SPARC fit, the a_0 -line and the flat curves are measurements against a law, independent of which object the action contracts.

2.1 New in v3 — eccentric orbits, and how sharp the prediction actually is

Real wide binaries are **not circular**, so §2’s identity is the wrong tool for them on its own, and this is the honest soft spot of v1 and v2. Off circles the eigenvector relation fails and a closure must be chosen — a prescription for reducing $\square_u u_\mu$ to the scalar argument of K . The repo’s prior verdict on that closure was **free, bounded**. Two things are new here.

(a) **The freedom is much smaller than “free” suggests — by two orders of magnitude.** The prior bracket was computed over time-weightings of $|a|^2$, which know nothing about $\square_u$. Using the action’s actual $\square_u$ instead collapses the spread. On radial infall onto $10^{12} M_\odot$ — the maximally non-circular case, i.e. a hard upper bound on eccentricity effects:

infall from	$ a ^2$ -weighting family	action $\square_u$ family	narrowing
300 kpc	$\nu =$ 1.170, 2.364, 7.837 — spread 570%	$\nu =$ 1.0004, 1.0062, 1.0797 — spread 7.9%	72×
100 kpc	spread 172%	spread 0.9%	189×

So the off-circular ambiguity in ν is $\lesssim 8\%$ in the worst case, not a factor of several. The s^3 prediction is correspondingly **sharp to that level for eccentric pairs**, which is what the Gaia DR4 shape test actually needs. This upgrades the prediction from quasi-linear to nearly sharp.

(b) **Two exact identities, and an honest limit.** Symbolically, at matched local acceleration a and radius r :

$$\text{circular: } \frac{\square_u u}{u} = -\Omega^2 = -\frac{a}{R}, \quad \text{radial infall: } \frac{\square_u u^r}{u^r} = \frac{2GM}{r^3} = +\frac{2a}{r}.$$

A factor 2 and an **opposite sign**. Stated against interest: the action does **not** eliminate the freedom, because $\square_u u$ is not parallel to u off circles — the spatial and time components differ by 8 to 5 orders along the infall — so a projection must still be chosen. The action narrows; it does not close.

3. The gate is shut in the window everyone measures — on the AC branch only

Everything in this section is conditional on the AC branch withdrawn in §2. On the DC branch the gate's gain at zero frequency is exactly 1, whatever ω_c is, so none of the suppression below applies and γ_v takes its ungated value ≈ 1.02 .

A pair of total mass M at separation s has $\Omega = (GM/s^3)^{1/2}$. For $M = 1.5 M_\odot$ at 10 kAU, $\Omega = 2.44 \times 10^{-13} \text{ s}^{-1}$ — **11× above the top of the whole ω_c window**. The gate is shut, $\text{Re } G = 0.005\text{--}0.008$, and $\gamma_v = 1.0004\text{--}1.0006$: within 0.04σ of Newton.

Two radii run in opposite directions with separation:

$$r_M = \sqrt{GM/a_0} \quad (\text{sub-}a_0 \text{ begins}), \quad r_{\text{gate}} = (GM/\omega_c^2)^{1/3} \quad (\text{gate opens}).$$

Since $\Omega \propto s^{-3/2}$ while $g \propto s^{-2}$, always $r_{\text{gate}} > r_M$ — the ratio is 4.54–7.76 across 0.5–3.0 $M_\odot$, both footings, both edges. **Between them a pair is already below a_0 but still gate-shut, so it must look Newtonian.** For 1.5 $M_\odot$ that dead zone runs ~10 to ~50–60 kAU.

Galaxies are untouched, which is what makes the gated reading admissible. Galactic orbital frequencies are $\Omega = v/r$, nine orders smaller. Across the Milky Way solar circle and outer disk, dwarfs, low-surface-brightness disks, massive spirals and the inner disk, the largest value found is $\Omega/\omega_c = 0.364$ ($\text{Re } G \geq 0.883$). Gate open in galaxies, shut in wide binaries: the radial acceleration relation and the baryonic Tully–Fisher relation are unaffected.

4. The cubic law (conditional on the AC branch) — and why the exponent counts poles

Deep in the gate-shut regime $\Omega \gg \omega_c$, so $\text{Re } G \rightarrow (\omega_c/\Omega)^2$; substituting $\Omega = (GM/s^3)^{1/2}$ gives $\text{Re } G \rightarrow (\omega_c^2/GM) s^3$, hence the boxed law. The measured local slope $d \ln(\gamma_v - 1)/d \ln s$ is **3.00** at 5 kAU, falling through the knee to 0.06 once saturated.

An n -pole gate gives $|G_n|^2 \rightarrow (\omega_c/\Omega)^{2n}$, so

$$\gamma_v - 1 \propto s^{3n} \quad \implies \quad n = p/3.$$

Verified numerically at $n = 1, 2, 3$ (slopes 3.00, 6.00, 8.99). **The measured exponent returns the gate's pole count.** The committed gate is one-pole and predicts exactly 3; 6 would mean two poles; anything not a multiple of 3 rules out the rational-pole form. (The identity $|G|^2 = \text{Re } G$ used above holds only at $n = 1$; for $n > 1$ the law uses $|G_n|^2$.)

The exponent is parameter-free even though the amplitude is not. ω_c is free (§1) and enters the amplitude — but it does **not** enter the exponent. That asymmetry is the whole reason this is worth measuring. Three mass powers follow from the same closed form and are recovered numerically: $r_{\text{gate}} \propto M^{0.3333}$, $r_{\text{efe}} \propto M^{0.5000}$, fixed-separation excess $\propto M^{-0.9935}$ (asymptotic; knee-contaminated to $M^{-0.857}$ by 30 kAU, so fit the local slope rather than assuming a global $1/M$).

5. The discriminant (conditional on the AC branch)

Every published contamination channel carries a fixed velocity scale, so its scaled relative velocity rises as $s^{1/2}$: chance alignment (El-Badry, Rix & Heintz 2021, via $R_{\text{chance align}}$), unresolved tertiaries (Peñarrubia 2021; Tyler, Green & Goodwin 2023), close-binary contamination. **Exponent 0.5 against 3.0.** Signal and background have different separation dependences, so a joint fit is well posed.

This matters because the amplitude axis is unusable: the two groups analysing the same public catalogue differ by $\Delta\gamma_v \approx 0.174$, about **2.1× the entire predicted boost**. No amplitude test survives that spread. A shape test does.

6. Why this waits for Gaia DR4 — now resolved, negatively

Version 1 said feasibility “cannot be settled without the catalogue in hand” and “may be fatal to a DR3 test.” We pulled the catalogue. It is fatal.

Because the excess grows as s^3 , nearly all the signal sits at large separation:

s [kAU]	γ_v	note
30	1.0125	the cut used by every published analysis
50	1.0392	—
100	1.0808	—
200	1.0932	catalogue’s reach

At the 30 kAU cut the predicted excess (1.2×10^{-2}) is an order of magnitude below the inter-group systematic — **the conventional window cannot test this**. And at our own recipe’s cuts ($d < 200$ pc, $R_{\text{chance align}} < 0.1$, $RUWE < 1.4$ both components, MS–MS), the El-Badry, Rix & Heintz (2021) Gaia eDR3 catalogue contains only

$$436 / 270 / 155 / 60 / 24 / 3 \text{ pairs across the } 30 \rightarrow 236 \text{ kAU bins,}$$

i.e. **364 clean pairs beyond 50 kAU** and ~ 70 beyond 100 kAU. The shape test needs ~ 2000 per bin for 5σ separation of $p = 3$ from $p = 0.5$ (~ 400 gives only $\sim 2.3\sigma$). Relaxing the distance cut does not rescue it: proper-motion velocity error scales as (distance $\times$ pm error) while the sample requirement scales as σ^2 , so raw counts at $d < 1000$ pc are $\sim 15\times$ larger but the usable gain is far smaller — and even at face value only 2 of 6 bins clear 2000, while a log-slope fit needs most bins populated.

Ground-based astrometry cannot substitute. The test needs $\sigma_{\text{pm}} \lesssim 0.02 \text{ mas yr}^{-1}$ at 200 pc; Rubin/LSST delivers $\sim 0.1\text{--}1.0$ over ten years. **Gaia DR4 is the only route.**

7. Pre-registration

Since the test cannot run until DR4, the prediction is frozen in the open beforehand. The pre-registration (`prep_2026/gaia_dr4_prep/PREREGISTRATION_DR4.md`, Amendment 1, 2026-07-27, SHA-256 `a309a502332b84ad521385b1c2031652849c9ce358c396f680f2308dd5ba1c13`) records:

- the frozen prediction $p = 3$ for the log-slope beyond r_{gate} ;

- $\gamma_v = 1.0004\text{--}1.0006$ inside 2–30 kAU, i.e. a Newtonian DR4 result in that window confirms this branch rather than refuting it — which the unamended pre-registration would have scored as a kill. **v6: the basis for this row is WITHDRAWN (§2).** It rests on the AC branch, and on the framework’s own DC branch the registered value is instead the ungated $\gamma_v \approx 1.02$. An **Amendment 6 is owed** to the frozen document and is the author’s to file before DR4; until it is filed, this row should be read as the AC-branch entry only, not as the framework’s prediction. Note also that withdrawing the AC branch removes only **one** of the frozen document’s two routes to a Newtonian reading — the locally-dragged frame branch reaches $\gamma_v - 1 \approx 1 \times 10^{-6}$ independently, with no gate — so the pre-registration’s falsifiability trap is **narrowed, not resolved**;
- the five falsifiers listed at the top of this note;
- and that DR4 γ_v constrains the ν +EFE+gate prescription, **not** the value of a_0 or Z . No outcome may be reported as measuring either.

8. What is not claimed

- **Not a test of a_0 or Z .** A confirmed s^3 rise establishes a frequency scale ω_c and a one-pole response. $Z = 5.7888$ and the conventional $2\pi = 6.2832$ differ by 7.87%, both lie inside the $\pm 16\%$ empirical a_0 box, and no arena examined here separates them. Against interest: matching Chae (2024b)’s central γ on this framework’s own kernel needs $a_0 \approx 1.9\times$ canonical, which at face value disfavors the Λ -tied coefficient specifically.
- **Nothing about whether dark matter exists.** A wide-binary result constrains the weak-field force law, not the matter content. $\gamma_v = 1$ is Newton’s prediction — shared by Λ CDM, by this framework’s own gated branch, and by modified-force-plus-particle hybrids.
- **a_0 ’s value, κ (equivalently Z), the sign $s = -1$ and ω_c are postulated, not derived.** New in v3: six independent classes of condition on the kernel’s own spectral structure were swept for one that would fix κ , and all six are closed — every committed condition sits at a scale-free point of $z = \square_u/a_0^2$ and is invariant under $a_0 \rightarrow \lambda a_0$.

New in v3 — the footing this paper carries is the one independently favoured, and credit for that question is not ours. The paper runs both a_0 footings throughout. The fork itself — whether a_0 couples to $cH(z)$ or to the dark-energy density $\sqrt{8\pi G\rho_\Lambda/3}$ — was posed and tested long before this work: **Gnedin (2008)**, The Redshift Evolution of the Tully–Fisher Relation as a Test of Modified Gravity (arXiv:0809.2790), states both options explicitly, tests them against high- z Tully–Fisher data to $z = 1.2$, and concludes that dark-energy coupling can be consistent while Hubble coupling requires unreasonable w — “this would appear to favor coupling to the dark energy density.” See also arXiv:0802.1526 on whether G and a_0 vary with epoch.

We add only an **independent check from a different epoch and different systematics**, which is worth recording because Gnedin explicitly flagged possible unaccounted systematics. Applying the two footings to the recombination plasma: since the acoustic acceleration is $a_{ac} \sim c_s H$, a rising $a_0 = cH(z)/Z$ makes H **cancel**, leaving $y = Z/\sqrt{3(1+R)} \approx 2.6\text{--}3.3$ at every epoch — so $\nu - 1 \approx 15\%$ throughout the entire acoustic era, not as a late correction. Propagated through $c_s^2 = c^2/[3(1+R)]$ over CAMB’s exact background (our sound-horizon integral reproduces CAMB’s own $r_* = 144.44$ Mpc to 0.00%), this shifts θ_* by $\approx 1.5\%$ against Planck’s 0.03% measurement — excluded. The constant, ρ_Λ -tied footing shifts θ_* by 3×10^{-7} and is consistent. **Same verdict as Gnedin (2008), from the CMB rather than from Tully–Fisher.** Stated as a limitation: this is a background-plus-recombination treatment, not a modification of the Boltzmann perturbation hierarchy, and a fully relativistic MOND theory (AeST; Skordis & Złośnik) already reproduces the

CMB — so the rigorous form of this question belongs to that theory, not to an $a_0(z)$ scaling argument.

Reproducibility. Every number is printed by committed scripts, all `exit 0`, no hard-coded verdicts, both footings and both ω_c edges throughout: `mi_dcac_branch_settled_2026.py` (5/5, §2 — **VERDICT WITHDRAWN 2026-07-31; superseded by** `mi_dcac_branch_verdict_withdrawn_2026.py`, 29/29, and not to be cited for its conclusion), `mi_wb_gate_fork_2026.py` (11/11, §3), `mi_wb_cubic_rise_2026.py` (10/10, §4), `mi_wb_exponent_pipeline_2026.py` (6/6, §4–5), `mi_wb_dr3_feasibility_2026.py` (5/5, §6), `mi_omegac_anchor_2026.py` (8/8, §1), `count_wb_elbadry2021.py` (catalogue counts). New in v3: `mi_kappa_spectral_reduction_2026.py` (§1, the kappa reduction), `mi_offcircular_closure_collapse_2026.py` (§2.1, the closure narrowing and the two identities), `mi_three_classes_2026.py` + `mi_thermal_class_nogo_2026.py` + `mi_bootstrap_circularity_2026.py` (§8, the six closed classes), `mi_cmb_camb_run_2026.py` (§8, the CAMB footing check). Repository: <https://github.com/carlzimmerman/zimmerman-formula>

Corrections carried from Version 1, recorded rather than quietly fixed: the 200-vs-30 kAU excess ratio is $7\times$, not the $>10\times$ first stated; the fixed-separation mass scaling is asymptotic ($M^{-0.9935}$ at 10 kAU) and knee-contaminated to $M^{-0.857}$ by 30 kAU; the operative sample requirement is the shape test’s ~ 2000 pairs/bin, not the ~ 265 implied by excess-detection alone; and ω_c is now declared a free parameter rather than described as a constrained window.

New in v3, and stated plainly: nothing in this version changes the s^3 prediction, its five falsifiers, or the frozen pre-registration. v3 sharpens the error budget (§2.1: eccentric-orbit closure now $\lesssim 8\%$ instead of a factor of several), simplifies the coefficient (§1: one rational $\kappa = 1/2$), and adds an independently-credited footing check (§8). No new observable, no new claim, and one additional negative result — six closed classes in the search for what fixes κ .

References

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